Real-Time Intrusion Detection for IoT Networks

NEURAL IDS — a deep-learning NIDS on CIC-IoT-2023

Rareș Marta

Faculty of Mathematics and Computer Science
Babeș-Bolyai University
Coordinator: *[name]*

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Motivation & purpose

Why it matters

- Billions of IoT devices, weak security — prime targets for DDoS, Mirai, scanning.
- Defenders need to flag malicious traffic *as it happens*, per flow.

Purpose

- A real-time intrusion-detection system that classifies network flows as *benign* or *attack* and surfaces alerts live.
- Delivered as a working end-to-end system — not just an offline model.

Outline

1. What an IDS is
2. Data & model — CIC-IoT-2023
3. System architecture
4. Demo
5. Results & limitations
6. Conclusions

What an IDS does — passive detect & alert

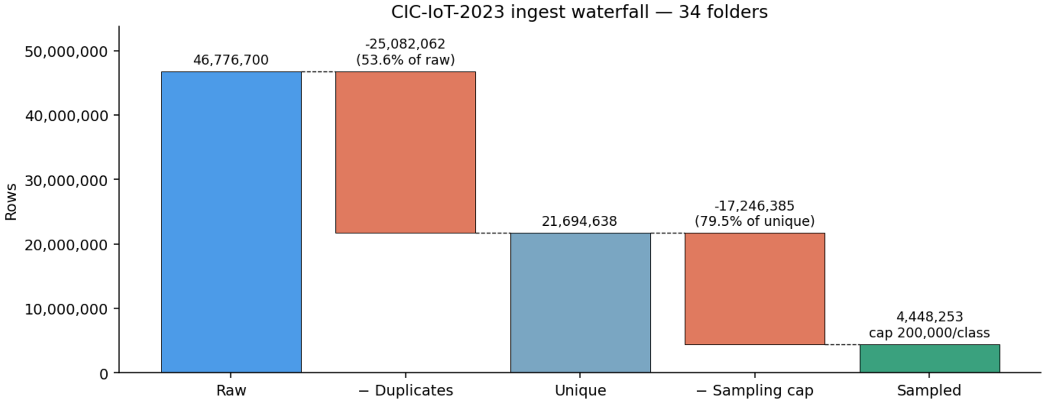
- Observes a copy of traffic — it *detects and alerts*, it does not block.
- Each 10-packet window between two hosts $\rightarrow$ 25 features $\rightarrow$ a calibrated verdict.

The dataset: CIC-IoT-2023

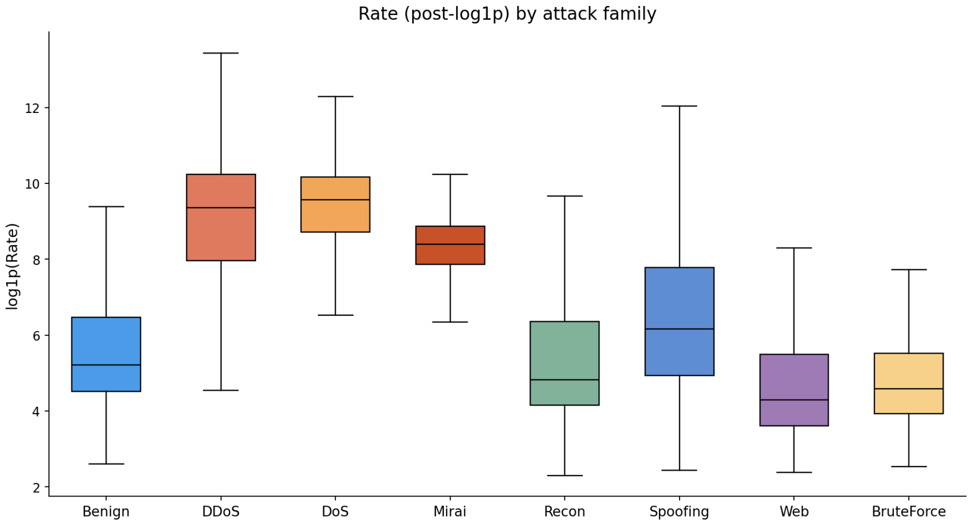
Canadian Institute for Cybersecurity
University of New Brunswick (2023)

- 46.7M labelled flows · 105 real IoT devices · smart-home testbed
- 33 attacks across 7 families · 39-feature public CSV (packet/flow metadata)
- **Caveat flagged:** per-class packet windows make count features a near-label proxy — treated as leakage, not signal.

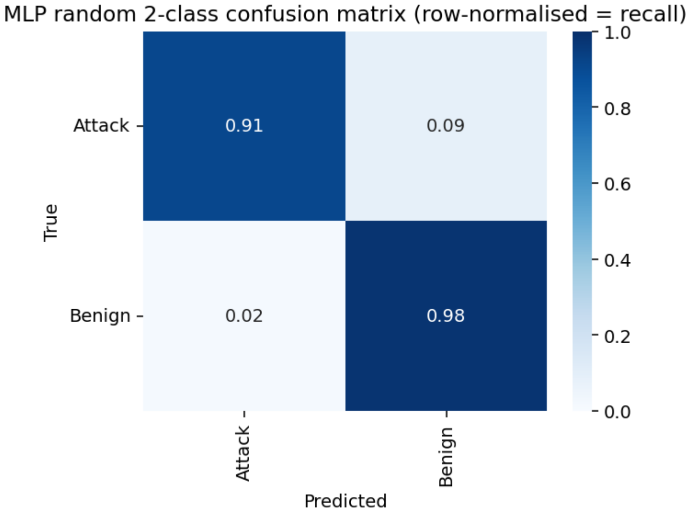
Cleaning 46.7M rows — dedup before sampling



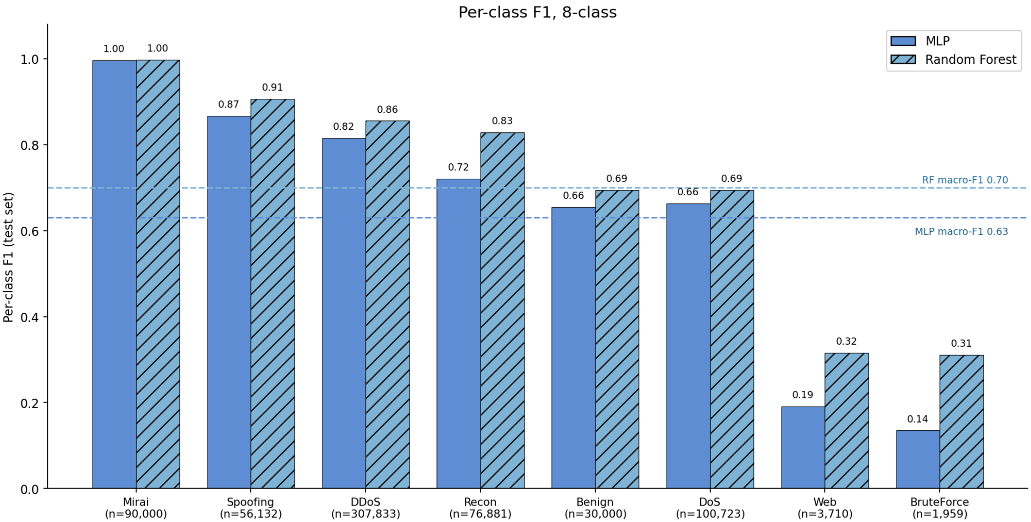
Feature separability — floods easy, web overlaps benign



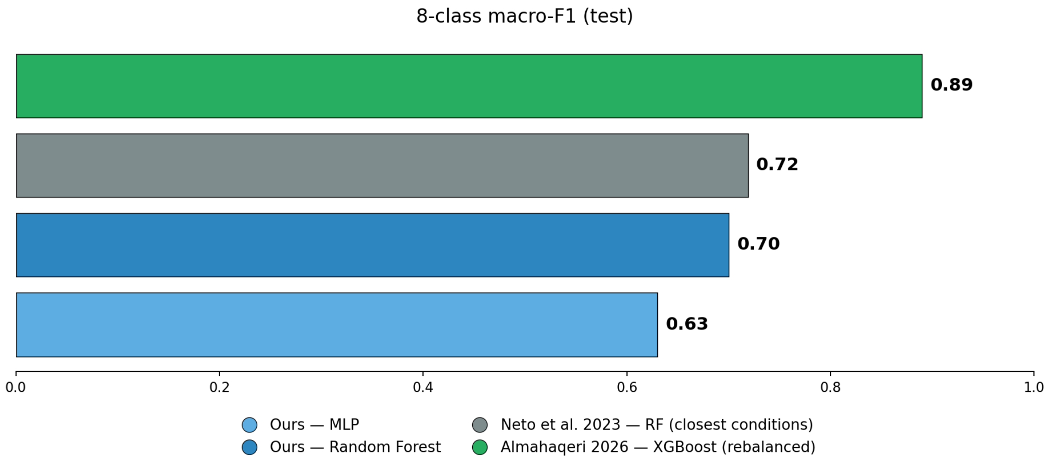
Binary gate — the primary task (test set)



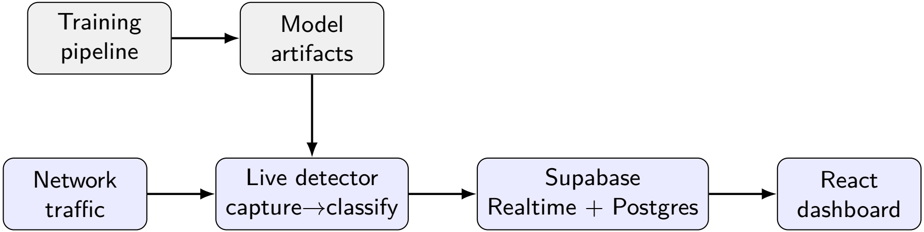
8-class: application-layer attacks collapse



Positioning — comparable results only



System architecture

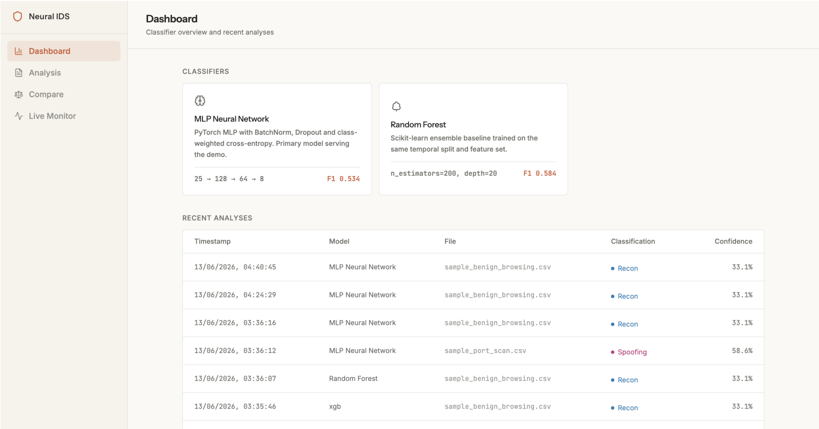


- Offline training emits artifacts; the live detector loads them and serves verdicts.
- Flows stream ephemerally over Realtime; incidents & stats persist in Postgres.

Technologies

Layer	Tooling
Model	PyTorch MLP · scikit-learn (RF, RobustScaler)
Data pipeline	Polars · Apache Parquet
Flow extraction	dpkt — custom CIC-window extractor
Detector service	FastAPI · Uvicorn
Backplane	Supabase (Postgres + Realtime)
Dashboard	React · Vite · Vercel
Live sensor	DigitalOcean droplet (Docker)

Demo — live monitoring



Scrolling flow stream · click a flow for detail · live traffic-rate chart · trigger an attack and watch it flag.

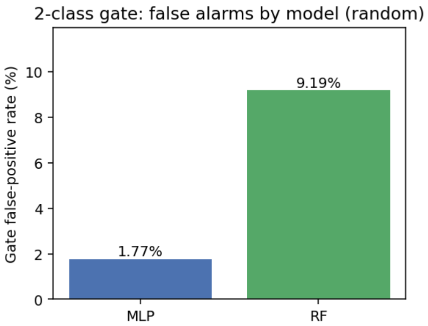
Testing — acceptance

End-to-end on captures whose correct labels are known in advance:

- Benign browsing $\rightarrow$ **98% Benign** (passes the 95% bar)
- SYN flood $\rightarrow$ **100% Attack**, mean confidence 99.6%, family $=$ DoS
- Port scan $\rightarrow$ family $=$ **Recon**

In-distribution the binary gate is reliable: **MLP false-alarm rate 1.77%**.

Limitation — out-of-distribution traffic



- In-distribution the gate over-flags only 1.77% of benign flows.
- On *real* internet traffic it over-flags heavily — the model's "Benign" was learned from smart-home IoT devices, a different distribution (domain shift).

Conclusions & future work

Delivered

- A defensible MLP baseline with tree comparison on CIC-IoT-2023.
- A working live, streaming IDS: capture $\rightarrow$ classify $\rightarrow$ dashboard.

Limitations

- Out-of-distribution over-flagging (domain shift); dataset windowing leakage; transport-only features miss web attacks; optimistic random split.

Future work

- Leakage-aware retraining; fine-tune on representative traffic; source allow-listing / baselining; line-rate deployment.